Yule's coefficient: $Q = \dfrac{(AB)(\alpha\beta) - (A\beta)(\alpha B)}{(AB)(\alpha\beta) + (A\beta)(\alpha B)}$

$$= \frac{180 \times 1055 - 445 \times 120}{180 \times 1055 + 445 \times 120}$$

$$= \frac{189900 - 53400}{189900 + 53400}$$

$$= \frac{136500}{243300}$$

$$= 0.561 > 0$$

Interpretation: There is a positive association between success in examination and attending coaching classes. Coaching class is useful for success in examination.

Remark: Consistency in the data using contingency table may be found as under.

Construct a 2 × 2 contingency table for the given information. If at least one of the cell frequencies is negative then there is inconsistency in the given data.

Example 4.8

Verify whether the given data: N = 100, (A) = 75, (B) = 60 and (AB) = 15 is consistent.

Solution:

The given information is presented in the following contingency table.

	B	β	Total
A	15	60	75
α	45	(-20)	25
Total	60	40	N = 100

Notice that $(\alpha\beta) = -20$

Interpretation: Since one of the cell frequencies is negative, the given data is "Inconsistent".

POINTS TO REMEMBER

- Correlation study is about finding the linear relationship between two variables. Correlation is not causation. Sometimes the correlation may be spurious.
- Correlation coefficient lies between –1 and +1.
- Pearson's correlation coefficient provides the type of relationship and intensity of relationship, for the data in ratio scale measure.
- Spearman's correlation measures the relationship between the two ordinal variables.
- Yule's coefficient of Association measures the association between two dichotomous attributes.

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